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# Final Project Report MATH 563

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## Abstract

The frustrating problem of *blurry* images has persisted since the invention of cameras. Multiple types of blurs occur for different reasons, such as *camera shake blur*, *motion blur*, or *out of focus blur*. These images are a nuisance, and pose challenges in many applications, such as image content recognition, surveillance, astronomy and many more [4]. Hence, reducing blur, or *deblurring* is crucial in solving the ubiquitous problem of blurred images in a society inundated with visual media. In this continually evolving research area, many methods have been proposed, broadly categorized into those addressing the *non-blind* and *blind* deblurring problems. This article will primarily focus on the former, where the blur kernel is known. We will discuss four algorithms designed to tackle this challenge, highlighting their strengths and weaknesses while providing illustrative examples.

## 1 Introduction

### 1.1 Notation

Throughout this paper, we will denote the *proximal operator* of  $f$  with parameter  $\lambda$  by

$$\mathbf{prox}_{\lambda f}(v) := \arg \min_x \left\{ f(x) + \frac{1}{2\lambda} \|x - v\|_2^2 \right\} \quad (1)$$

The *indicator function* for a convex set  $C$  will be defined as

$$\delta_C(x) := \begin{cases} 0, & \text{if } x \in C \\ +\infty, & \text{if } x \notin C \end{cases} \quad (2)$$

We also denote the *Euclidean Projection* on a set  $C$  by

$$\mathbf{P}_C(v) := \arg \min_{x \in C} \|x - v\|_2 \quad (3)$$

### 1.2 Preliminaries

We first note an important relation between the proximal and subdifferential operators.

**Proposition 1.** Given a function  $f \in \Gamma$ , we have

$$\mathbf{prox}_{\lambda f} = (I + \lambda \partial f)^{-1}, \quad (4)$$

where  $(I + \lambda \partial f)^{-1}$ , for  $\lambda > 0$ , is called the **resolvent** of the operator  $\partial f$ . In particular, the proximal operator is the resolvent of the subdifferential operator [3].

**Definition 1** (Maximal Monotone operator). A monotone operator is **maximal** if its graph is not properly contained in the graph of any other monotone operator. If  $A$  is monotone (resp. maximal monotone) then so are  $A^{-1}$  and  $\lambda A$  if  $\lambda > 0$ .

The next proposition, given in [2], will be useful in the construction of the optimizing algorithms.

**Proposition 2.** If  $F$  is maximal monotone, then  $(I + \lambda F)^{-1}(\hat{x})$  exists and is unique for all  $\hat{x} \in \mathbb{R}^n$ . The value  $x = (I + \lambda F)^{-1}(\hat{x})$  of the resolvent is the unique solution of the monotone inclusion

$$\hat{x} \in x + \lambda F(x) \quad (5)$$

**Proposition 3** (Proximal operator splitting). If  $f$  is separable across two variables, so  $f(x, y) = \varphi(x) + \psi(y)$ , then

$$\text{prox}_f(x, y) = (\text{prox}_\varphi(x), \text{prox}_\psi(y)) \quad (6)$$

Thus, evaluating the proximal operator of a separable function reduces to evaluating the proximal operators for each of the separable parts, which can be done independently [3].

**Proposition 4.** (Proximal operator of norms) If  $f = \|\cdot\|$  is a general norm, then  $f^*$  is the indicator function on the unit ball for the dual norm  $\|\cdot\|_*$ , namely

$$\|\cdot\|^* = \delta_{\|\cdot\|_* \leq 1}$$

where  $\|\cdot\|_*$  is given by

$$\|v\|_* = \sup_x \{\langle v, x \rangle \mid \|x\| \leq 1\}$$

In particular, the *Moreau Decomposition Theorem* gives that

$$\text{prox}_{\|\cdot\|}(v) = v - \mathbf{P}_{\{x: \|x\|_* \leq 1\}}(v)$$

### 1.3 Derivations of the proximal maps

We provide a quick derivation of the proximal operators of the L1 norm, the L2 norm, the iso norm, and the box prox. By the Moreau decomposition, we naturally obtain the proximal operators of the Fenchel conjugates. Let  $b \in \mathbb{R}^d$  be fixed.

- Let  $g : y \mapsto t\|y - b\|_1$ . We have

$$\text{prox}_g(y) := \arg \min_u \left\{ t\|u - b\|_1 + \frac{1}{2}\|y - u\|_2^2 \right\} = \begin{cases} b & y - b \in [-t, t] \\ y - \text{sign}(y - b) \cdot t & \text{otherwise} \end{cases}$$

We consider first-order optimality conditions, with the following cases:

- If  $u - b > 0$ , then

$$t - y + u = 0 \quad \Leftrightarrow \quad u = y - t$$

- If  $u - b < 0$ , then

$$-t - y + u = 0 \quad \Leftrightarrow \quad u = y + t$$

- If  $u - b = 0$ , then  $u = b$  under which by optimality conditions we have

$$0 \in t \cdot [-1, 1] - y + b \quad \Leftrightarrow \quad y - b \in [-t, t]$$

- Now consider  $g : y \mapsto t\|y - b\|_2^2$ . This  $g$  is continuously differentiable so we can take its gradient directly and set it to zero. That is, the steps are:

$$\frac{\partial}{\partial u} \left( t\|u - b\|_2^2 + \frac{1}{2}\|y - u\|_2^2 \right) = 2t(u - b) - y + u = u(2t + 1) - y - 2tb$$

Therefore

$$\text{prox}_g(y) = \frac{y + 2tb}{2t + 1}$$

- For  $g : (y_1, y_2) \mapsto t\gamma \|(y_1, y_2)\|_{\text{iso}}$  where

$$\|(x, y)\|_{\text{iso}} = \sum_{k=1}^n (x_k^2 + y_k^2)^{1/2}$$

To compute the proximal operator of  $g$ , we use Moreau decomposition (7.5.8). The Fenchel conjugate of  $g$  is  $g^*(y_1, y_2) = t\gamma \max_{1 \leq k \leq n} (y_{1k}^2 + y_{2k}^2)^{1/2}$  where  $y_1, y_2 \in \mathbb{R}^n$  (simplified after elementary conjugacy operations). So we have

$$\mathbf{prox}_g(y_1, y_2) = (y_1, y_2) - \mathbf{prox}_{g^*}(y_1, y_2)$$

where

$$\mathbf{prox}_{g^*}(y_1, y_2) = \arg \min_{(u, v)} \left\{ t\gamma \max_{1 \leq k \leq n} (u_k^2 + v_k^2)^{1/2} + \frac{1}{2} \|(y_1, y_2) - (u, v)\|_2^2 \right\}$$

Remark that  $(u_k^2 + v_k^2)^{1/2} = \|(u_k, v_k)\|_2$  for each  $k$ . The Fenchel conjugate of the L2 norm is itself (HW3). We also have that for any norm  $\|\cdot\|$ , the Fenchel conjugate  $\|\cdot\|^*$  is the indicator of the unit ball in the dual norm  $\delta_{\|\cdot\|^* \leq 1}$ . The proximal mapping of the indicator of the unit ball is the projection onto the unit ball for points not in the unit ball. Hence

$$\mathbf{prox}_{g^*}(y_1, y_2) = t\gamma \frac{(y_1, y_2)}{\|(y_1, y_2)\|_2}$$

as long as  $(y_{1k}^2 + y_{2k}^2)^{1/2} > 0$ . If  $(y_{1k}^2 + y_{2k}^2)^{1/2} = 0$ , then  $y_{1k} = y_{2k} = 0$  so  $(u, v) = (0, 0)$ . So altogether, for  $k = 1, \dots, n$

$$\mathbf{prox}_g(y_{1k}, y_{2k}) = \begin{cases} \left(1 - \frac{t\gamma}{(y_{1k}^2 + y_{2k}^2)^{1/2}}\right)(y_{1k}, y_{2k}) & (y_{1k}^2 + y_{2k}^2)^{1/2} > 0 \\ 0 & \text{otherwise} \end{cases}$$

- For the box prox,  $f : x \mapsto t\delta_S(x)$  where  $S := \{x : x \in [0, 1]\}$ , the proximal operator is the projection. That is,

$$\begin{aligned} \mathbf{prox}_f(x) &= \arg \min_u \{t\delta_S(u) + \frac{1}{2}\|x - u\|_2^2\} \\ &= t \cdot \arg \min_u \{\delta_S(u) + \frac{1}{2t}\|x - u\|_2^2\} \\ &= \arg \min_{u \in S} \frac{1}{2}\|x - u\|_2^2 \end{aligned}$$

## 2 Algorithms

The algorithms we present are used to solve the non-blind image deblurring problem, which is a convex optimization problem of the form

$$\begin{aligned} &\min_{x, y} f(x) + g(y) \\ &\text{subject to } Ax = y, \end{aligned} \tag{7}$$

where  $f, g \in \Gamma$  have inexpensive prox-operators. We are given a vectorized, blurred image  $b$ , along with the kernel  $K$  used to blur it using the model

$$Kx + \eta = b, \tag{8}$$

where  $x$  is the original image, and  $\eta$  the added noise. Our goal is to recover the original image  $x$  from the blurred and noisy image  $b$ . We formulate the problem in two ways, namely, one which uses the  $L^1$  norm, and the other the  $L^2$  norm. First,

$$\min_x \|Kx - b\|_1 + \delta_S(x) + \gamma \|Dx\|_{\text{iso}}, \tag{9}$$

and second,

$$\min_x \|Kx - b\|_2^2 + \delta_S(x) + \gamma \|Dx\|_{\text{iso}}, \quad (10)$$

where  $S = \{x \mid 0 \leq x \leq 1\}$ , so that  $\delta_S$  aims to restrict the range of allowed pixels.

The algorithms are based on a splitting scheme to facilitate computation of the resolvents of the maximal monotone operators. As mentioned in a bit more detail in the project prompt, we have from the fixed point iteration scheme

$$x^k = (I + t\mathcal{F})^{-1}(x^{k-1}) \quad k = 1, 2, \dots$$

where  $\mathcal{F}$  is a maximal monotone operator. Splitting algorithms decompose the problem into

$$\begin{aligned} x^k &= (I + t\mathcal{A})^{-1}(z^{k-1}) \\ y^k &= (I + t\mathcal{B})^{-1}(2x^k - z^{k-1}) \\ z^k &= z^{k-1} \rho(y^k - x^k) \end{aligned}$$

where  $t > 0$  is the step size and  $\rho \in (0, 2)$  is a relaxation parameter. Splitting algorithms are used en lieu of a classic fixed point iteration due to expensive computation costs.

## 2.1 Primal Douglas-Rachford splitting (Spingarn's method)

This first algorithm, which we will refer to by PDRS, was invented in the 1950s as a method to solve the heat equation [1]. Per (9), let

$$\begin{aligned} f(x) &= \delta_S(x) \\ g(x) &= \|Kx - b\|_1 + \gamma \|Dx\|_{\text{iso}} \end{aligned} \quad (11)$$

and consider the minimization problem

$$\min_{x,y} f(x) + g(y) + \delta_0 \left( \begin{bmatrix} A & -I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \right) \quad (12)$$

with the optimality condition

$$0 \in \partial f(x) + \partial g(y) + \partial \delta_0 \left( \begin{bmatrix} A & -I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \right) =: \mathcal{F}(x, y)$$

As stated in the literature [2],  $\mathcal{F}$  is the subdifferential of closed, convex functions, hence is a maximal monotone operator. Thus, by (5), the zeros of  $\mathcal{F}$  are the fixed points of the resolvent of  $\mathcal{F}$ . Therefore, if we can find the fixed points of  $\mathcal{F}$ , we'll have found the optimal points. Unfortunately, this particular problem is too computationally expensive, so the strategy is to split  $\mathcal{F}$  into two maximal monotone operators  $\mathcal{A}$  and  $\mathcal{B}$  which have inexpensive proximal operators. Finding the fixed points of  $\mathcal{A} + \mathcal{B}$  will be equivalent to finding that of  $\mathcal{F}$ . Define  $\mathcal{A}$  and  $\mathcal{B}$  as,

$$\begin{aligned} \mathcal{A}(x, y) &= \partial f(x) + \partial g(y) = \partial(\delta_S(x)) + \partial(\|Ky - b\|_1 + \gamma \|Dy\|_{\text{iso}}) \\ \mathcal{B}(x, y) &= \partial \delta_0 \left( \begin{bmatrix} K \\ D \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \right) \end{aligned} \quad (13)$$

We now focus on finding the resolvents of  $\mathcal{A}$  and  $\mathcal{B}$ ,

$$\begin{aligned} J_{\mathcal{A}} &= (I + \lambda \mathcal{A})^{-1} = \mathbf{prox}_{\lambda(f+g)}(x, y) \quad [\text{by (4)}] \\ &= (\mathbf{prox}_{\lambda f}(x), \mathbf{prox}_{\lambda g}(y)) \quad [\text{by Prop. 3}] \end{aligned} \quad (14)$$

where

$$\mathbf{prox}_{\lambda f}(x) = \mathbf{prox}_{\lambda \delta_S}(x) = \arg \min_u \frac{1}{2\lambda} \|x - u\|_2^2 + \delta_S(x) = \arg \min_{u \in S} \frac{1}{2\lambda} \|x - u\|_2^2 = \mathbf{P}_S(x), \quad (15)$$

which is the euclidean projection operator onto  $S$ . It can be computed element-wise by

$$(\mathbf{P}_S(x))_i = \begin{cases} 0, & \text{if } x_i < 0 \\ x_i, & \text{if } x_i \in S = \max\{0, \min\{x_i, 1\}\} \\ 1, & \text{if } x_i > 1 \end{cases} \quad (16)$$

Now, we observe that  $g(y)$  described in (11) can be written in a different form by letting

$$g(y_1, y_2, y_3) = \|y_1 - b\|_1 + \gamma \|(y_2, y_3)\|_{\text{iso}} \quad (17)$$

subject to

$$y_1 = Ky \quad \text{and} \quad \begin{bmatrix} y_2 \\ y_3 \end{bmatrix} = Dy \iff \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} K \\ D \end{bmatrix} y$$

Then, by Proposition 3, notice that the proximal operator of  $g$  may be obtained by splitting its components into

$$\varphi(y_1) = \|y_1 - b\|_1 \quad \text{and} \quad \psi(y_2, y_3) = \gamma \|(y_2, y_3)\|_{\text{iso}}$$

which gives

$$\mathbf{prox}_{\lambda g}(y_1, y_2, y_3) = (\mathbf{prox}_{\lambda \varphi}(y_1), \mathbf{prox}_{\lambda \psi}(y_2, y_3)) \quad (18)$$

We know that the proximal operator of the  $L^1$  norm is the soft-thresholding operator  $T_\lambda$ , so  $\mathbf{prox}_{\lambda \varphi}(y_1)$  is given element-wise as follows,

$$(\mathbf{prox}_{\lambda \varphi}(y_1))_i = (\mathbf{prox}_{\lambda \|\cdot\|_1}(y_1 - b))_i = T_\lambda((y_1 - b)_i) := \begin{cases} (y_1 - b)_i - \lambda, & \text{if } (y_1 - b)_i > \lambda \\ (y_1 - b)_i + \lambda, & \text{if } (y_1 - b)_i < -\lambda \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

As a result of Proposition 4,  $\mathbf{prox}_{\lambda \psi}(y_2, y_3)$ , is given element-wise by

$$(\mathbf{prox}_{\lambda \psi}(y_2, y_3))_i = \mathbf{prox}_{\gamma \lambda \|\cdot\|_{\text{iso}}}(y_2, y_3)_i = \alpha_i((y_2)_i, (y_3)_i),$$

where,

$$\alpha_i = \begin{cases} 1 - \frac{\gamma \lambda}{\sqrt{(y_2)_i^2 + (y_3)_i^2}}, & \text{if } \sqrt{(y_2)_i^2 + (y_3)_i^2} > \gamma \lambda \\ 0, & \text{otherwise.} \end{cases}$$

As for the resolvent of  $\mathcal{B}$ , it corresponds to the projection  $(x, y)$  of a vector  $(\bar{x}, \bar{y})$  onto the subspace  $\{(x, y) \mid [K \ D]^T x = y\}$ , expressed as

$$J_{\mathcal{B}} = (I + \lambda \mathcal{B})^{-1} = \left( x = \left( I + [K \ D] \begin{bmatrix} K \\ D \end{bmatrix} \right)^{-1} \left( \bar{x} + \begin{bmatrix} K \\ D \end{bmatrix} \bar{y} \right), y = \begin{bmatrix} K \\ D \end{bmatrix} x \right)$$

The algorithm is included in Appendix A.1.

## 2.2 Primal-Dual Douglas Rachford Splitting

This second method uses both optimality conditions of the primal problem and the dual problem for the "split". The method takes advantage of the inexpensive computation of the prox-operators of  $f$  and  $g$ , and the fact that our matrix  $A$  makes it easy to compute linear equations.

In the algorithm, we consider the dual problem,

$$\text{maximize } -f^*(-A^T z) - g^*(z) \quad (20)$$

with the optimality conditions for both the primal and the dual problem

$$0 \in \begin{bmatrix} 0 & 0 & A^T & I \\ 0 & 0 & -I & 0 \\ -A & I & 0 & 0 \\ -I & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} + \begin{bmatrix} 0 \\ \partial g(y) \\ 0 \\ \partial f^*(w) \end{bmatrix}$$

In the above,  $w$  is the dual variable and  $z$  is the dual multiplier for the constraint  $y = Ax$ , thus the dual problem is subject to

$$A^T z + w = 0$$

We reduce this to a smaller system by using the relationship  $\partial g^* = (\partial g)^{-1}$ , which follows from the fact that  $g$  is closed and convex, and by the properties of the subgradient and conjugate. We can eliminate  $y$  by using the above equality that  $\partial g^* = (\partial g)^{-1}$ , which would leave us with

$$0 \in \begin{bmatrix} 0 & A^T & I \\ -A & 0 & 0 \\ -I & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ z \\ w \end{bmatrix} + \begin{bmatrix} 0 \\ \partial g^*(z) \\ \partial f^*(w) \end{bmatrix}$$

We can further reduce this by eliminating  $w$ , using a similar equality that  $\partial f = (\partial f^*)^{-1}$ , we get

$$0 \in \begin{bmatrix} 0 & A^T \\ -A & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} \partial f(x) \\ \partial g^*(z) \end{bmatrix} \quad (21)$$

We can now split the monotone operator we have above to a sum of two monotone operators given as

$$\mathcal{A}(x, z) = \begin{bmatrix} \partial f(x) \\ \partial g^*(x) \end{bmatrix} \quad \mathcal{B}(x, z) = \begin{bmatrix} 0 & A^T \\ -A & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} \quad (22)$$

Similar to the previous algorithm, we now find the resolvents of  $\mathcal{A}$  and  $\mathcal{B}$ .

As seen before with the resolvent of  $\mathcal{A}$ , the value  $(x, y) = (I + At)^{-1}(\hat{x}, \hat{z})$  is given by

$$x = \mathbf{prox}_{tf}(\hat{x}) \quad z = \mathbf{prox}_{tg^*}(\hat{z})$$

For  $z$ , we can apply moreau decomposition to get an alternate expression dealing with the conjugate  $g^*$ , thus we have,

$$z = \hat{z} - t \mathbf{prox}_{t^{-1}g}(\hat{z}/t)$$

Simplifying the above, we are left with,

$$z = \hat{z} - \mathbf{prox}_{gt}(\hat{z})$$

This will be a similar computation to the prox operator in algorithm 1, but we have right scalar multiplication in this case.

The resolvent for  $\mathcal{B}$  is a linear mapping, so considering resolvents for skew-symmetric linear operators

$$\begin{aligned} (I + t\mathcal{B})^{-1} &= \begin{bmatrix} I & tA^T \\ -tA & I \end{bmatrix}^{-1} = \frac{1}{I - (tA^T)(-tA)} \begin{bmatrix} I & -tA^T \\ tA & I \end{bmatrix} = \frac{1}{I + t^2A^TA} \begin{bmatrix} I & -tA^T \\ tA & I \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} + \frac{1}{I + t^2A^TA} \begin{bmatrix} I \\ tA \end{bmatrix} \begin{bmatrix} I & -tA^T \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I + t^2A^TA)^{-1} \begin{bmatrix} I & -tA^T \end{bmatrix}^T \end{aligned}$$

Letting  $(w, v)$  be the value of the resolvent of  $\mathcal{B}$ ,

$$\begin{aligned} \begin{bmatrix} w \\ v \end{bmatrix} &= \begin{bmatrix} I & tA^T \\ -tA & I \end{bmatrix}^{-1} \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} = \left( \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I + t^2A^TA)^{-1} \begin{bmatrix} I & -tA^T \end{bmatrix}^T \right) \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I + t^2A^TA)^{-1} \begin{bmatrix} I & -tA^T \end{bmatrix}^T \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ \hat{z} \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I + t^2A^TA)^{-1} \begin{bmatrix} I & -tA^T \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ \hat{z} \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I + t^2A^TA)^{-1} (\hat{x} - tA^T\hat{z}) \\ &= \begin{bmatrix} 0 \\ \hat{z} \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} \frac{\hat{x} - tA^T\hat{z}}{(I + t^2A^TA)} \end{aligned}$$

$$= \begin{bmatrix} 0 \\ \hat{z} \end{bmatrix} + \begin{bmatrix} \frac{\hat{x} - tA^T \hat{z}}{(I + t^2 A^T A)} \\ tA \left( \frac{\hat{x} - tA^T \hat{z}}{(I + t^2 A^T A)} \right) \end{bmatrix} = \begin{bmatrix} \frac{\hat{x} - tA^T \hat{z}}{(I + t^2 A^T A)} \\ \hat{z} + tA \left( \frac{\hat{x} - tA^T \hat{z}}{(I + t^2 A^T A)} \right) \end{bmatrix}$$

Thus we have the solution for both  $w$ ,  $v$ ,

$$w = \frac{\hat{x} - tA^T \hat{z}}{(I + t^2 A^T A)} = \frac{\hat{x}}{(I + t^2 A^T A)} - \frac{tA^T \hat{z}}{(I + t^2 A^T A)} \quad (23)$$

$$v = \hat{z} + tA \left( \frac{\hat{x} - tA^T \hat{z}}{(I + t^2 A^T A)} \right) = \hat{z} + \frac{tA\hat{x}}{(I + t^2 A^T A)} - \frac{t^2 AA^T \hat{z}}{(I + t^2 A^T A)} \quad (24)$$

Notice in the algorithm, we will let  $\hat{x} = 2x^k - p^{k-1}$  and  $\hat{z} = 2z^k - p^{k-1}$ . The Primal-Dual Douglas Rachford splitting algorithm is included in Appendix A.2.

This algorithm resembles the first algorithm in section 2.1. They are both splitting algorithm that use the prox-operators of  $f$  and  $g$ . The algorithms differ in the fact that the Primal-Dual Douglas Rachford splitting algorithm uses both the primal and dual optimality conditions, rather than uniquely using the primal as done in section 2.1. This means that we need to use Moreau decomposition for the conjugate of the prox-operator of  $g$ .

### 2.3 ADMM (Dual Douglas-Rachford)

The dual problem from above

$$\text{maximize}_y -f^*(-A^T y) - g^*(y) \quad (25)$$

has the equivalent form

$$\text{minimize}_z -f^*(-A^T z) - g^*(z) := h(z) + k(z)$$

and it is said that applying the Douglas-Rachford algorithm to this dual problem results in the alternating direction method of multipliers (ADMM) algorithm. That is, ADMM uses the Douglas-Rachford iteration on the dual problem. Optimality is guaranteed when  $0 \in \partial h(z) + \partial k(z)$  in which we have two maximal monotone operators with resolvents  $\text{prox}_{th}(z)$  and  $\text{prox}_{tk}(z)$ .

ADMM, like the previous algorithms, is a splitting algorithm. Unlike the previous two algorithms, it has different optimality conditions, that is it uses solely optimality conditions of the dual problem. Apart from this, it relies, as the other algorithms do, on the Douglas-Rachford iteration, with a different update ordering as mentioned in the project prompt. The proximal operators are fundamentally the same despite different inputs and despite using the inverse of the step size  $t^{-1}$ . This is can be interpreted by the reliance of the Lagrangian on functions  $f$  and  $g$ . Another difference is the use of the step size  $t$  to update  $w^k$  and  $z^k$ , whereas previously somewhat analogous updates used the relaxation parameter  $\rho$ . It turns out the  $\rho$  is still used, but hidden within the update steps.

A form of the Douglas-Rachford iteration on the dual can be derived to be the following, as given in the project prompt:

$$\begin{aligned} \hat{x}^k &= \arg \min_x \left( f(x) + (z^{k-1} - w^{k-1})^T Ax + \frac{t}{2} \|Ax\|^2 \right) \\ v^k &= z^{k-1} - w^{k-1} + tA\hat{x}^k \\ \hat{y}^k &= \arg \min_y \left( g(y) - (v^k + w^{k-1})^T y + \frac{t}{2} \|y\|^2 \right) \\ z^k &= v^k + w^{k-1} - t\hat{y}^k \\ w^k &= w^{k-1} + \rho v^k + (1 - \rho)z^{k-1} - z^k. \end{aligned}$$

In particular, we are interested in the general problem

$$\begin{aligned} &\min_{u,y} f(u) + g(y) \\ &\text{subject to } \begin{bmatrix} I \\ A \end{bmatrix} x + \begin{bmatrix} -I & 0 \\ 0 & -I \end{bmatrix} \begin{bmatrix} u \\ y \end{bmatrix} = 0 \end{aligned}$$

for which we have an augmented Lagrangian given as

$$L(x, u, y, w, z) = f(u) + g(y) + w^T(x - u) + z^T(Ax - y) + \frac{t}{2} (\|x - u\|^2 + \|Ax - y\|^2)$$

which makes no difference in the above formulation as we alternate minimization over  $x$  and minimization over  $(u, y)$ , so that they are constants in alternation. The pseudo code can be seen in [Appendix A.3](#).

## 2.4 Chambolle-Pock

The Chambolle-Pock method considers the generic optimization problem

$$\begin{aligned} & \min_{x, y} f(x) + g(y) \\ & \text{subject to } Ax = y, \end{aligned}$$

by finding the solution of

$$\min_x \max_y \langle Ax, y \rangle + f(x) - g^*(y). \quad (26)$$

It is mentioned that in our case, the solution turns out to be the Lagrangian of our generic optimization problem.

The Chambolle-Pock algorithm stands out as the odd one out among the four algorithms. It does not use a relaxation parameter  $\rho$ , but two step sizes  $t$  and  $s$  which depend on the size of the operators in the problem, as mentioned in the project prompt. Furthermore, this algorithm does not require solving a system of linear equations involving  $I + A^T A$ . We only need vector-matrix multiplications of  $A$  and  $A^T$ , hence giving a time complexity that is not necessarily the same as the previous algorithms.

Chambolle-Pock is not based on an "optimality condition" as the previous three, but properly chosen step sizes  $s$  and  $t$  that will guarantee convergence. In particular, as described in the project prompt, by choosing the step sizes  $s$  and  $t$  such that  $st(\|A\|_2^2) < 1$ , we can ensure the following convergence:

$$\frac{\|y^k - \hat{y}\|_2^2}{2s} + \frac{\|x^k - \hat{x}\|_2^2}{2t} \leq C \left( \frac{\|y^0 - \hat{y}\|_2^2}{2s} + \frac{\|x^0 - \hat{x}\|_2^2}{2t} \right) \quad (27)$$

In summary, because the sequence of iterates  $\{(x^k, y^k)\}_k$  is bounded, by extracting subsequences, convergence to a fixed point is guaranteed.

Despite these differences, it is said that the Chambolle-Pock algorithm is at its core, a form of preconditioned ADMM. Unpacking this formulation, and with the help of the equations

$$y^k = (I + s\partial g^*)^{-1}(y^{k-1} + sAz^{k-1}) \quad z^k = 2x^k - x^{k-1}$$

due to the addition of

$$\frac{1}{2} \left\langle \left( \frac{1}{s} - tAA^T \right) (y - y^{k-1}), y - y^{k-1} \right\rangle$$

to the augmented Lagrangian at the  $k$ -th iteration, we see that the updates of  $y^k$  and  $z^k$  are exactly two of the steps of the algorithm described above. The algorithm proceeds as below. It utilizes the proximal operators of  $g^*$  and  $f$ , resembling the primal-dual formulation. The algorithm is included in [appendix A.4](#).

## 3 Tuning and Results

In this section, we discuss on the hyperparameters and the algorithm results. To gauge the performance of the algorithms, on top of the obtained image, we use an objective function of the form  $x \mapsto \|x - x_0\|_2^2$  for some objective  $x_0$ . This function resembles the mean-squared error. It turns out using as objective the blurred image  $b$  suffices for decent results, as we will in fact use the function

$$x \mapsto \|Kx - b\|_2^2$$

as objective function, where  $K$  is the blurring matrix and  $b$  is the blurred image. It is evident that if the blurring matrix blurs the image in a way that does not resemble the blurred image  $b$  at all, this objective function will not serve its purpose, hence the need of choosing a best  $K$ . Such a  $K$  will naturally depend on the blurred image. So in fact, we do not want to pick a best  $K$ , but merely one that gives acceptable results for any blurred image  $b$ .

Other than the blurring kernel  $K$ , we are also interested in the effects of the additive noise  $\eta$ . We will try different choices of additive noise and compare the results.

Apart from the blurring kernel  $K$  and the additive noise  $\eta$ , we are also interested in the effects of the step size  $t > 0$  and the relaxation parameter  $\rho \in (0, 2)$ . Their effects will be recorded in their subsection. A final parameter of interest is  $\gamma > 0$  that appears in the expression  $\gamma \|Dx\|_{iso}$ . As suggested by the project prompt, this  $\gamma$  will depend on the choice of problem, either L1 or L2.

After appropriately-chosen hyperparameters have been set, we will present the results of the four algorithms, in particular, the obtained, hopefully deblurred, images. We will compare these results and finally, leave concluding remarks.

We implemented a grid search function to check the various values of a single parameter with all the others fixed. Then, we created a separate file and tested various combinations of parameters to find the best combination, ( $6^3$  total to be precise), to be precise, all the resulting plots are left in the results folder of the matlab code.

Table 1: Default Parameters Used

| Hyperparameter  | In-line Name i. | Value |
|-----------------|-----------------|-------|
| Kernel          | kernel          |       |
| Noise           |                 |       |
| Total Iteration | maxiter         | 500   |
| gamma L1        | gamma1          | 0.1   |
| gamma L2        | gamma2          | 0.02  |
| t               | tprimaldr       |       |
|                 | rhoprimaldr     |       |
|                 | tprimaldualdr   |       |
|                 | rhoprimaldualdr |       |
|                 | tadmm           |       |
|                 | rhoadmm         |       |
|                 | scp             |       |
|                 | tcp             |       |

### 3.1 Hyperparameter tuning

#### Blurring kernel and noise

We denote the blurring kernel by  $K$  and the noise by  $\eta$ . We will consider the following blurring kernels  $K$

- Gaussian blurring
- Motion blurring

and the following additive noise  $\eta$

- Poisson noise
- Salt and pepper noise

add some others if u want to impress a certain prof

#### Gamma

#### Step size and relaxation parameter

### 3.2 Results

### 3.3 Conclusion

### Acknowledgments

Please include a list of each person (with student ID) in your group and write a paragraph **for each person** with exactly what they did. Note that the Acknowledgment section does not count towards the page limit.

Example Acknowledgments would look like:

**Alexandre St-Aubin** (Student ID: 261115785) We ball.

**Jonathan Campana** (Student ID: 260985613) We ball.

**Elliot Forcier-Poirier** (Student ID: ) We ball.

**Edmund Wu** (Student ID: 261033501) We ball.

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### A.1 Primal Douglas-Rachford Splitting Algorithm

```

Input           : Blurred image  $b$ , step size  $t$ , relaxation parameter  $\rho$ 
Output          : Deblurred image  $x$ 
Initialize      :  $(z_1^0, z_2^0)$ 
for  $k = 1, 2, \dots$ , do
     $x^k \leftarrow \text{prox}_{tf}(z_1^{k-1})$ ;           // Performs resolvent of  $\mathcal{A}$ 
     $y^k \leftarrow \text{prox}_{tg}(z_2^{k-1})$ ;
     $u^k \leftarrow (I + A^T A)^{-1} (2x^k - z_1^{k-1} + A^T (2y^k - z_2^{k-1}))$ ; // Performs resolvent of  $\mathcal{B}$ 
     $v^k \leftarrow A(u^k)$ ;
     $z_1^k \leftarrow z_1^{k-1} + \rho(u^k - x^k)$ ;
     $z_2^k \leftarrow z_2^{k-1} + \rho(v^k - y^k)$ ;
end
return  $x\text{Sol} = \text{prox}_{tf}(z^k)$ ;

```

---

**Algorithm 2:** Primal-Dual Douglas-Rachford Splitting

|                                                                                                                                                                        |                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Input</b>                                                                                                                                                           | : Blurred image $b$ , step size $t$ , relaxation parameter $\rho$ |
| <b>Output</b>                                                                                                                                                          | : Deblurred image $x$                                             |
| <b>Initialize</b>                                                                                                                                                      | : $p^0, q^0$                                                      |
| <b>1 for</b> $k = 1, 2, \dots$ , <b>do</b>                                                                                                                             |                                                                   |
| 2 $x^k \leftarrow \text{prox}_{tf}(p_1^{k-1})$ ;                                                                                                                       | // Performs resolvent of $\mathcal{A}$                            |
| 3 $z^k \leftarrow \text{prox}_{ig^*}(q^{k-1})$ ;                                                                                                                       |                                                                   |
| 4 $\begin{bmatrix} w^k \\ v^k \end{bmatrix} = \begin{bmatrix} I & tA^T \\ -tA & I \end{bmatrix}^{-1} \begin{bmatrix} 2x^k - p^{k-1} \\ 2z^k - q^{k-1} \end{bmatrix}$ ; | // Performs resolvent of $\mathcal{B}$                            |
| 5 $p^k \leftarrow p^{k-1} + \rho(w^k - x^k)$ ;                                                                                                                         |                                                                   |
| 6 $q^k \leftarrow q^{k-1} + \rho(v^k - z^k)$ ;                                                                                                                         |                                                                   |
| <b>7 end</b>                                                                                                                                                           |                                                                   |
| <b>8 return</b> $x\text{Sol} = \text{prox}_{tf}(p^k)$ ;                                                                                                                |                                                                   |

### A.3 ADMM Algorithm

---

**Algorithm 3:** Alternating Direction Method of Multipliers (ADMM)

---

**Input** : Blurred image  $b$ , step size  $t$ , relaxation parameter  $\rho$   
**Output** : Deblurred image  $x$   
**Initialize** :  $x^0, u^0, y^0, w^0, z^0$

```

1 for  $k = 1, 2, \dots$ , do
2    $x^k \leftarrow (I + A^T A)^{-1} (u^{k-1} + A^T y^{k-1} - \frac{1}{t}(w^{k-1} + A^T z^{k-1}))$ ;
3    $u^k \leftarrow \text{prox}_{t^{-1}f}(\rho x^k + (1 - \rho)u^{k-1} + w^{k-1}/t)$ ;
4    $y^k \leftarrow \text{prox}_{t^{-1}g}(\rho A x^k + (1 - \rho)y^{k-1} + z^{k-1}/t)$ ;
5    $w^k \leftarrow w^{k-1} + t(x^k - u^k)$ ;
6    $z^k \leftarrow z^{k-1} + t(A x^k - y^k)$ ;
7 end
8 return  $x\text{Sol} = (I + A^T A)^{-1} (u^k + A^T y^k - \frac{1}{t}(w^k + A^T z^k))$ ;

```

---

### A.4 Chambolle-Pock Algorithm

---

**Algorithm 4:** Chambolle-Pock Algorithm

---

**Input** : Blurred image  $b$ , step sizes  $t, s$   
**Output** : Deblurred image  $x$   
**Initialize** :  $x^0, y^0, z^0$

```

1 for  $k = 1, 2, \dots$ , do
2    $y^k = \text{prox}_{sg^*}(y^{k-1} + s A z^{k-1})$ ;
3    $x^k = \text{prox}_{tf}(x^{k-1} - t A^T y^k)$ ;
4    $z^k = 2x^k - x^{k-1}$ ;
5 end
6 return  $x\text{Sol} = x^k$ ;

```

---